KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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PROJECTS

Small-Agent QLoRA 2026

Python, Qwen3.5-9B, QLoRA / PEFT, Transformers, llama.cpp

- Fine-tuned a Qwen3.5-9B tool agent with QLoRA, improving task success from **34.5% to 62.0%** on a frozen 200-task evaluation.
- Turned failure traces into corrective training data and used ablation tests to reject a training variant that hurt held-out performance.
- Built the agent runtime with search/read/Python tools, duplicate-call blocking, timeouts, and trace logging; GAIA100 accuracy improved from **9% to 18%** over the base model.

Semantic Text-to-SQL 2026

Python, FastAPI, LangGraph, SQLite, Qwen3.5-9B

- Built a local Text-to-SQL service that grounds on the database schema, validates generated SQL, repairs failures, and executes queries read-only.
- On 100 frozen BIRD DEV queries, the guarded path improved execution success from **75% to 90%** and execution-match accuracy from **33% to 39%**.
- Added before/after regression checks that recovered **6 wrong answers with 0 correct-to-wrong regressions**; dropped a semantic-review pass after it reduced accuracy.

EXPERIENCE

Kahap Inc Aug 2022 – Aug 2026

Software Developer (Part-time)

- Led development and production support for a feedback platform from initial build through deployment and ongoing maintenance.
- Built submission, discussion, admin, and notification workflows that moved the process into one centralized system.
- Maintained a Laravel/Vue.js career platform and handled queues, scheduled jobs, SSL, database issues, and Linux server migrations.

Pac Com Solution Jun 2021 – Jan 2022

Full-Stack Developer Intern *Malaysia*

- Built and maintained Laravel/Vue.js features for a production POS system, including third-party API integrations and frontend/backend fixes.

RESEARCH

M.S. Thesis: Implicit Intent in Workplace Dialogue 2025 – 2026

NLP, retrieval, workplace-role and context graphs

- Built a labeled workplace-dialogue dataset with **20,455 utterances** across **7 boards** for implicit intent and listener-side interpretation.
- Combined role and context graphs with retrieval, improving target-only macro-F1 from **0.625 to 0.689** on **401 test utterances**.

TECHNICAL SKILLS

AI / LLM:	LLM agents & tool calling, QLoRA / PEFT, Hugging Face Transformers, RAG, retrieval & LLM evaluation, LangGraph
Programming / Backend:	Python, SQL, PHP, JavaScript, FastAPI, Laravel, REST APIs
Data / Systems:	MySQL, SQLite, Linux, Git, GitHub Actions, pytest, llama.cpp
Frontend:	Vue.js, React
Languages:	Chinese (Native/Bilingual), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University Sep 2024 – Jun 2026

M.S., Information Systems & Applications | GPA: 4.09/4.3 *Taiwan*

Chung Hua University Sep 2022 – Jun 2024

B.S., Computer Science & Information Engineering | GPA: 3.9/4.0 *Taiwan*